

(a) FUTURE PREDICTION



OBSERVED ONLY

V-JEPA encoder

context features
FROZEN BASE

(b) LATENT TRANSPORT

FactorJEPA predictor



$$\hat{Y}_{t+\Delta} = Y_L + Y_A + Y_I$$

Latent transport

$\mathcal{T}_\omega$
tokens $\rightarrow$ spatial tensor
 $\mathbb{R}^{m \times d_J} \rightarrow \mathbb{R}^{h \times w \times d_C}$

TRAINABLE

(c) RGB DECODING

Cosmos decoder

$\mathcal{G}_{\psi_C}$
latent $\rightarrow$ RGB
COSMOS INIT.

(d) QUALITY AUDIT



Quality

PSNR $\uparrow$ SSIM $\uparrow$
LPIPS $\downarrow$ EPE $\downarrow$
Agent F1 $\uparrow$
HELD-OUT

ORACLE CONTROL · SEPARATING DECODER AND FORECAST ERROR



Target encoder

oracle latent
 Y^*

Shared $\mathcal{T}_\omega + \mathcal{G}_{\psi_C}$

shared weights
NO EXTRA CAPACITY



Decoder error

$d(\tilde{x}^*, x^*)$
rendering ceiling

Forecast gap

$d(\hat{x}, \tilde{x}^*)$
forecast-induced error

End-to-end quality

$d(\hat{x}, x^*)$
forecast + rendering

End-to-end error $\leq$ forecast-induced gap + oracle rendering error